

1.1

$$\begin{aligned} \textcircled{a} \quad \|u+v\|^2 &= \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} = \|u\|^2 + \|v\|^2 + 2\operatorname{Re}\langle u, v \rangle \end{aligned}$$

$$\begin{aligned} \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u, u \rangle + \langle v, v \rangle - \langle u, v \rangle - \langle v, u \rangle = \\ &= \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \overline{\langle u, v \rangle} = \|u\|^2 + \|v\|^2 - 2\operatorname{Re}\langle u, v \rangle \end{aligned}$$

poznaťi je sme lineárnu v oboch složkách, "skoro" symetriu a identitu $z + \bar{z} = 2\operatorname{Re}(z)$

$$\|u+v\|^2 - \|u-v\|^2 = 2\operatorname{Re}\langle u, v \rangle - (-2\operatorname{Re}\langle u, v \rangle)$$

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\begin{aligned} \textcircled{b} \quad \|u-iv\|^2 &= \langle u, u \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle = \\ &= \|u\|^2 + \|v\|^2 - i\langle u, v \rangle + i\langle v, u \rangle = \\ &= \|u\|^2 + \|v\|^2 - i\langle u, v \rangle + i\overline{\langle u, v \rangle} = \|u\|^2 + \|v\|^2 + 2\operatorname{Im}\langle u, v \rangle \end{aligned}$$

$$\begin{aligned} \|u+iv\|^2 &= \|u\|^2 + \|v\|^2 + i\langle u, v \rangle - i\langle v, u \rangle = \\ &= \|u\|^2 + \|v\|^2 - (-i\langle u, v \rangle + i\overline{\langle u, v \rangle}) = \|u\|^2 + \|v\|^2 - 2\operatorname{Im}\langle u, v \rangle \end{aligned}$$

~~poznaťi je sme~~ poznáme vlastnosti skal. součinu a identitu $-iz + i\bar{z} = 2\operatorname{Im}(z)$

$$\|u-iv\|^2 - \|u+iv\|^2 = 4\operatorname{Im}\langle u, v \rangle$$

$$\operatorname{Im}\langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$